

Exploring the psychological puzzle of travelling: A Journey Through Your Thoughts

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ABSTRACT

Daily commuting, often overlooked as routine, is a rich terrain for exploring the psychological dimensions of travel. This abstract delves into the intricate interplay between our minds and the journeys we undertake each day. Even though we might not think much about it, our daily commutes—whether by walking, biking, driving, or taking public transport—have a big impact on how we feel. This abstract looks at the psychology behind these everyday travels. As we travel our familiar routes, our minds can drift between focusing on the journey and thinking about other things. Sometimes, we find moments of peace or use the time to think about our lives. But commuting isn't always smooth sailing. Dealing with traffic jams, delays, or crowded buses can make us feel stressed or annoyed. Our daily commutes might seem routine, but they play a big role in how we think and feel. By understanding the psychology behind our everyday travels, we can learn more about ourselves and how to make the most of our journeys.

INTRODUCTION

Every single day, as people start their day, many of us go through something that seems pretty normal but actually affects us a lot—the daily commute. This is when we travel to work, school, or wherever we need to go. Even though it might seem boring or just something we have to do, it actually has a big impact on how we feel and think. In big cities, like Karachi with tall buildings and lots of traffic, commuting can be stressful. We have to deal with crowded streets and busy roads, which can make us feel frustrated or annoyed. But at the same time, we learn how to handle these challenges and become more resilient. On the other hand, in quieter areas like suburbs or countryside, the commute might be more peaceful. We might drive through beautiful landscapes and have time to think or relax. However, it can still have its own difficulties, like long distances or bad weather. No matter where we commute, it's not just about getting from one place to another—it's also about how it makes us feel. In this exploration, we'll look at how daily commuting affects our emotions, thoughts, and connection to the world around us. By understanding this, we can learn how to make the most of our everyday travels.

Scope of research

To explore the psychological aspects of daily commuting and understand their impact on individuals' mental well-being. To shed light on the complex interplay between daily travel and psychological health.

- **Importance of Daily Commuting:** Daily commuting is a widespread phenomenon that affects millions of individuals worldwide. Understanding the psychological dynamics of commuting is essential due to its pervasive nature and potential impact on mental well-being.
- **Limited Understanding:** Despite the prevalence of daily commuting, there is a gap in our understanding of its psychological aspects. While previous research has explored the physical and environmental aspects of commuting, there is a need to delve deeper into the thoughts, emotions, and experiences of commuters.
- **Impact on Mental Well-being:** Daily commuting has been associated with various mental health outcomes, including stress, anxiety, and depression. By examining commuters' thoughts, emotions, and coping strategies, this study aims to shed light on the relationship between daily travel and psychological health.
- **By identifying factors that contribute to both positive and negative psychological outcomes during commuting, this research can inform strategies to enhance the commuting experience and support commuters' mental health.**

Overall, the rationale for this study lies in the importance of understanding the psychological aspects of daily commuting, the potential impact on mental well-being, and the opportunity for interventions to improve the commuting experience and support commuters' mental health.

Rationale of study

Importance of Daily Commuting:

1. Daily commuting is a widespread phenomenon that affects millions of individuals worldwide. Understanding the psychological dynamics of commuting is essential due to its pervasive nature and potential impact on mental well-being.
2. **Limited Understanding:** Despite the prevalence of daily commuting, there is a gap in our understanding of its psychological aspects. While previous research has explored the physical and environmental aspects of commuting, there is a need to delve deeper into the thoughts, emotions, and experiences of commuters.
3. **Impact on Mental Well-being:** Daily commuting has been associated with various mental health outcomes, including stress, anxiety, and depression. However, the specific mechanisms through which commuting influences mental well-being remain unclear. By examining commuters' thoughts, emotions, and coping strategies, this study aims to shed light on the complex relationship between daily travel and psychological health.
4. **Potential for Intervention:** Understanding the psychological dynamics of commuting has practical implications for interventions aimed at promoting mental well-being. By identifying factors that contribute to both positive and negative psychological outcomes during commuting, this research can inform strategies to enhance the commuting experience and support commuters' mental health.
5. **Contribution to Knowledge:** This study contributes to the growing body of literature on transportation psychology by providing insights into the psychological puzzle of daily commuting. By unraveling this puzzle, we can gain a deeper understanding of the psychological dynamics of commuting and its implications for mental well-being.

Overall, the rationale for this study lies in the importance of understanding the psychological aspects of daily commuting, the potential impact on mental well-being, and the opportunity for interventions to improve the commuting experience and support commuters' mental health

Addressing issues

To address issues related to commuting, independent variables can be introduced to influence or change certain factors affecting the commute experience. These variables can be manipulated to observe their effects on the overall commuting experience. Here's how independent variables can help solve commuting issues.

Research Question

How do different modes of transportation (e.g., car, public transit, walking, cycling) influence commuters' mental well-being? What factors contribute to both positive and negative psychological outcomes during daily commuting?

Research Objectives

The primary objective of this research is to comprehensively examine the influence of the reliability of transportation options on various aspects of commuter experience, mode choice behavior, and overall quality of life in urban environments. Specifically, the study aims to achieve the advance understanding of the complex relationship between the reliability of transportation options and commuter satisfaction, mode choice behavior, and quality of life in urban environments. Ultimately, the research aim at creating more resilient, efficient, and equitable transportation systems that meet the diverse needs of urban commuters.

Theoretical Framework and Conceptual Framework

The conceptual framework for this study is grounded in transportation psychology which considers the psychological factors in shaping individuals' well-being. The framework integrates key concepts and variables related to daily commuting and mental well-being, providing a theoretical basis for understanding the psychological dynamics of commuting.

Commuting Experiences:

1. **Mode of Transportation:** This variable encompasses different modes of commuting, including car, public transit, walking, and cycling. Each mode presents unique challenges and opportunities for commuters, influencing their mental well-being.
2. **Commute Duration:** The amount of time spent commuting is a crucial factor that can impact individuals' stress levels, mood, and overall well-being.
3. **Convenience and Accessibility:** The ease of access to transportation options and the convenience of commuting arrangements play a role in commuters' satisfaction and mental well-being.
4. **Safety and Reliability:** Perceptions of safety and reliability in transportation systems influence commuters' feelings of security and control, which in turn affect their mental well-being.

Psychological Responses:

1. **Stress, Anxiety, and Depression:** These psychological outcomes are commonly associated with daily commuting, particularly in situations of congestion, delays, or overcrowding.

2. Coping Mechanisms: Commuters employ various coping strategies to manage stress and improve their commuting experience by distraction techniques.
3. Loneliness and Isolation: Transportation barriers and lack of social interaction opportunities may contribute to feelings of loneliness and isolation among commuters, impacting their mental health.

Mental Well-being Outcomes:

1. Positive and Negative Affect: Commuting experiences can elicit both positive and negative emotions, which in turn influence individuals' overall mental well-being.
2. Quality of Life: The cumulative impact of commuting experiences on individuals' quality of life reflects their subjective well-being and satisfaction with their daily routine.

By integrating these concepts and variables, the conceptual framework provides a holistic understanding of the psychological puzzle of daily commuting and its implications for mental well-being. It guides the analysis and interpretation of data, informing interventions and policies aimed at promoting healthier commuting experiences and supporting commuters' mental health.

Research Hypothesis

Null Hypothesis (H0): There is no significant relationship between daily commuting experiences and mental well-being.

Alternative Hypothesis (H1): There is a significant relationship between daily commuting experiences and mental well-being.

Specific Hypotheses:

Mode of Transportation:

H0: There is no significant difference in mental well-being outcomes among individuals commuting by different modes of transportation.

H1: Commuters using public transit experience higher levels of stress and lower mental well-being compared to those using private vehicles or active transportation modes.

Safety and Reliability:

H0: Perceived safety and reliability of transportation options do not significantly impact mental well-being.

H1: Commuters who perceive their transportation options as safe and reliable report better mental well-being outcomes.

Testing these hypotheses will provide insights into the specific relationships between daily commuting experiences and mental well-being, guiding interventions and policies aimed at promoting healthier commuting habits and supporting commuters' mental health.

Significance of Research

The significance of research on the psychological puzzle of daily commuting lies in its potential to address critical issues affecting individuals' mental well-being and overall quality of life. Here are some key points highlighting the significance of this research:

Understanding Commuting Experiences: Daily commuting is a routine activity for many individuals, but its psychological impact is often overlooked. By exploring commuters' thoughts, emotions, and experiences, this research sheds light on the subjective aspects of commuting that influence mental well-being.

Addressing Mental Health Challenges: Commuting has been associated with various mental health issues, including stress, anxiety, and depression. Understanding the psychological dynamics of commuting can help identify risk factors and protective factors that contribute to mental well-being, informing targeted interventions and support services.

Supporting Sustainable Transportation: Promoting alternatives to solo car commuting, such as public transit, walking, and cycling, not only benefits the environment but also has potential mental health benefits. Understanding the psychological factors that influence mode choice can inform efforts to encourage sustainable transportation behaviors.

Improving Quality of Life: Commuting is a significant part of daily life for many people, and its impact extends beyond the time spent traveling. By addressing the psychological aspects of commuting, this research has the potential to improve overall quality of life for commuters, enhancing their well-being and satisfaction with daily routines.

Overall, research on the psychological puzzle of daily commuting is significant for its potential to inform policies, interventions, and practices that promote mental well-being, enhance the commuting experience, and create healthier, happier communities

LITERATURE REVIEW

The psychological aspects of daily commuting have garnered increasing attention in transportation psychology research, reflecting the recognition of commuting as a significant component of modern life with potential implications for individuals' mental well-being. This literature review provides a detailed examination of key findings from existing studies, organized by themes relevant to the psychological puzzle of daily commuting. Research consistently demonstrates a relationship between longer commute durations and negative mental health outcomes. Novaco (1997) found that commuters with longer travel times reported higher levels of stress, frustration, and fatigue compared to those with shorter commutes. This relationship is attributed to factors such as time pressure, fatigue, and reduced leisure time (Evans & Wener, 2007; Friman et al., 2017). Ettema et al. (2012) conducted a longitudinal study and found that increases in commute duration were associated with decreases in overall life satisfaction and well-being, highlighting the enduring impact of long commutes on individuals' psychological health.

Commuting by public transit has been a focus of research due to its association with higher stress levels compared to other modes of transportation. Hjorthol et al. (2012) found that public transit commuters reported greater feelings of stress and dissatisfaction with their commute, attributed to factors such as crowding, delays, and lack of control over the travel environment. In contrast, active modes of transportation, such as walking and cycling, have been associated with positive mental health outcomes. Stigell et al. (2016) found that individuals who walked or cycled to work reported lower levels of stress and higher life satisfaction compared to car commuters. The physical activity and sense of autonomy associated with active commuting contribute to its positive effects on mental well-being (Guell et al., 2018).

Commuters employ a variety of coping mechanisms to manage the stress and discomfort associated with commuting. Anable et al. (2005) identified several coping strategies, including listening to music, reading, engaging in mindfulness practices, and seeking social interaction during the commute. These coping strategies serve to mitigate the negative effects of commuting on mental well-being by providing a sense of control and relaxation (Molloy et al., 2019). Chatterjee and Mohan (2015) conducted a qualitative study and found that commuters who engaged in mindfulness practices reported decreased stress levels and increased feelings of relaxation during their commute. Mindfulness-based interventions have emerged as promising approaches for enhancing commuters' well-being and coping with the challenges of daily travel.

Environmental factors, such as noise, air pollution, and urban design, can influence commuters' perceptions of their surroundings and contribute to stress and discomfort during travel (Evans &

Wener, 2007; Gatersleben & Uzzell, 2007). Commuters exposed to high levels of environmental stressors may experience negative mental health outcomes, including increased anxiety and irritability. Contextual factors, including job satisfaction, work-life balance, and social support networks, interact with commuting experiences to influence overall mental well-being (Fisher et al., 2018; van den Berg et al., 2020). Commuters with supportive social networks and flexible work arrangements may experience lower levels of stress and greater satisfaction with their commute. Commuting can significantly impact family dynamics and relationships. Research by Özkan and Lajunen (2005) suggests that long commutes may lead to decreased family time and increased strain on familial relationships, particularly for individuals with significant caregiving responsibilities. Conversely, shorter commutes or shared transportation modes may facilitate greater family cohesion and satisfaction (Mokhtarian et al., 2013). Understanding the familial repercussions of commuting is crucial for addressing its broader social and psychological implications.

The commute itself represents a transitional period between home and work, wherein individuals may experience heightened levels of stress and anticipation regarding their impending workday (Rosenbloom, 2003). Research indicates that the quality of the commute, including factors like predictability and perceived control, can influence individuals' work-related stress levels and overall job satisfaction (de Vos et al., 2017). Exploring the interface between commuting and work-related stress offers insights into strategies for optimizing both commuter well-being and workplace productivity. The cognitive demands of commuting, such as navigating traffic or public transportation schedules, can impact individuals' cognitive functioning and mental acuity. Studies have shown that prolonged exposure to traffic congestion is associated with cognitive decline and impairments in attention and memory (Yuan et al., 2018). Understanding the cognitive implications of commuting sheds light on potential interventions to mitigate cognitive fatigue and enhance commuter cognitive resilience.

Commuting represents a unique opportunity for social interaction and community engagement. Research by Nieuwenhuis and Van Kann (2017) suggests that shared transportation modes, such as carpooling or public transit, facilitate social connectedness and interpersonal relationships among commuters. Conversely, solitary commutes, such as driving alone, may exacerbate feelings of isolation and loneliness (Carstensen et al., 2011). Exploring the social dimensions of commuting elucidates its role in fostering or hindering social cohesion and community well-being.

Promoting sustainable transportation modes, such as public transit and active transportation, not only reduces environmental impact but also offers potential benefits for commuter mental health. Research by Wang et al. (2020) indicates that individuals who use sustainable transportation modes report higher levels of life satisfaction and well-being compared to car-dependent commuters. Understanding the intersections between commuting behavior, sustainability, and mental well-being informs efforts to promote eco-friendly transportation policies and initiatives.

The daily commute can significantly impact individuals' sleep quality and overall well-being. Research by Buxton et al. (2012) suggests that longer commute times are associated with shorter sleep durations and poorer sleep quality, which can have detrimental effects on cognitive function, mood, and overall health. Understanding the relationship between commuting and sleep quality offers insights into interventions aimed at improving commuter well-being through strategies such as flexible work schedules and telecommuting options. The experience of commuting may also intersect with mental health stigma, affecting individuals' willingness to

seek support or disclose mental health challenges. Research by LaMontagne et al. (2014) indicates that individuals who experience stress or anxiety related to commuting may be hesitant to discuss these issues due to perceived stigma or fear of negative consequences in the workplace. Addressing mental health stigma within the context of commuting is essential for creating supportive environments where individuals feel comfortable seeking assistance when needed. The amount of leisure time available to individuals after commuting plays a crucial role in their overall well-being and life satisfaction. Research by Kikuchi et al. (2015) suggests that longer commute durations are associated with reduced leisure time, leading to lower levels of satisfaction and increased feelings of time pressure. Understanding the impact of commuting on leisure time satisfaction highlights the importance of policies and interventions aimed at reducing commute times and promoting work-life balance.

Gender dynamics can also influence commuting experiences and psychological outcomes. Research by Schwanen et al. (2017) suggests that women may experience different stressors and challenges during their commute compared to men, such as safety concerns and caregiving responsibilities. Addressing gender disparities in commuting experiences is essential for promoting gender equity and ensuring that transportation policies and infrastructure meet the diverse needs of all commuters. Commuting and Technology Use: The advent of technology has transformed commuting experiences and introduced new opportunities for enhancing mental well-being during travel. Research by Thomée et al. (2011) suggests that individuals may use smartphones and other devices to engage in activities such as listening to music, reading, or communicating with others, thereby mitigating boredom and stress during the commute. Understanding the role of technology in commuting behavior offers insights into the design of interventions and applications aimed at improving commuter satisfaction and well-being. By incorporating these additional points, the literature review provides a comprehensive examination of the diverse psychological dimensions of daily commuting, offering valuable insights into the complex interplay between commuting experiences and mental well-being across various contexts and populations.

Beyond individual mental health outcomes, commuting patterns can also impact broader community well-being. Research by Handy et al. (2006) suggests that commuting behaviors, such as reliance on single-occupancy vehicles versus public transit, can have significant environmental and social consequences for communities, including congestion, air pollution, and social isolation. Understanding the community-level effects of commuting sheds light on the importance of sustainable transportation policies and urban planning initiatives aimed at promoting healthier and more connected communities. The daily commute represents a unique space where individuals negotiate their identities and social roles. Research by Urry (2000) suggests that commuting experiences contribute to the construction of individual identities, shaping perceptions of self and others. Factors such as transportation mode choice, route selection, and interactions with fellow commuters can influence individuals' sense of identity and belonging. Exploring the role of commuting in identity formation offers insights into the complex interplay between transportation behavior and psychosocial development.

Despite the challenges associated with commuting, individuals may develop resilience strategies to navigate and cope with the daily stressors of travel. Research by Kloep and Hendry (2007) suggests that commuters may employ cognitive reframing, problem-solving, and social support-seeking strategies to adapt to the demands of commuting and maintain psychological well-being. Understanding the resilience factors associated with commuting offers opportunities for

interventions aimed at fostering adaptive coping mechanisms and enhancing commuter resilience in the face of adversity. Cultural norms and values can influence individuals' attitudes and behaviors towards commuting, shaping their psychological experiences and well-being. Research by Kitamura et al. (1997) suggests that cultural factors, such as collectivism versus individualism, may influence commuting preferences and perceptions of commute quality. Understanding cultural differences in commuting behavior and attitudes offers insights into the diverse ways in which individuals from different cultural backgrounds navigate the challenges of daily travel. Transitions in life stages, such as starting a new job, moving to a new location, or becoming a parent, can significantly impact individuals' commuting experiences and psychological well-being. Research by Friman et al. (2019) suggests that life transitions may alter commuting patterns, preferences, and stress levels, as individuals adapt to new circumstances and responsibilities. Understanding the intersection of commuting and life transitions provides valuable insights into the dynamic nature of commuting experiences across the lifespan.

Overall, the literature underscores the complex relationship between daily commuting experiences and psychological well-being. While commuting is often associated with negative outcomes such as stress and anxiety, there are also opportunities for positive experiences and coping strategies that enhance well-being. Understanding the multifaceted nature of commuting is essential for developing interventions and policies that promote healthier commuting habits and support commuters' mental well-being.

RESEARCH METHODOLOGY

Mixed-methods study involving surveys and interviews with commuters.

Research design;

Surveys will collect quantitative data on commute duration, mode of transportation, stress levels, mood, and coping mechanisms. Qualitative interviews will provide deeper insights into commuters' experiences, allowing for the exploration of themes such as frustration, boredom, social interaction, and relaxation during the commute. Diverse sample of commuters, including individuals from different demographic backgrounds and geographic locations, to ensure comprehensive insights. Reliance on self-reported data, which may be subject to biases and inaccuracies. Potential difficulty in capturing the full range of commuters' experiences and emotions due to the complexity of daily commuting. Limited generalizability of findings to specific geographic regions or commuting contexts. This research will contribute to a deeper understanding of the psychological dynamics of daily commuting and its impact on mental well-being.

Research approach:

The research approach for studying the psychological puzzle of daily commuting involves a mixed-methods approach, which integrates both quantitative and qualitative methods to provide a comprehensive understanding of the research phenomenon.

Unit of analysis:

Each participant in the study represents a distinct unit of analysis, providing data on their commuting experiences, perceptions, and mental well-being outcomes. The focus is on understanding the psychological dynamics of commuting at the individual level, exploring how various factors such as mode of transportation influence commuters' mental well-being. By examining the experiences of individual commuters, the research aims to uncover patterns,

themes, and insights that contribute to a broader understanding of the relationship between daily commuting and psychological health.

Sampling:

The sampling process is crucial for ensuring the generalizability and validity of the research findings. This approach ensures that every member of the population has an equal chance of being included in the sample, enhancing the representativeness of the study. A larger sample size increases the precision and reliability of the study. Ensure confidentiality and anonymity of participant data to protect privacy.

Data collection tool:

The data collection tool for this research on the psychological puzzle of daily commuting consists of a structured survey questionnaire. It is used to collect quantitative data from a large sample of commuters. The questionnaire includes closed-ended questions with predefined response options, allowing for efficient data collection and statistical analysis. Key components of the survey questionnaire include: data on participants' age, gender, education, including mode of transportation, commute duration. Along with it, Mental Well-being Measures: including stress, anxiety, life satisfaction, and overall psychological health.

Data collection procedure:

Primary data was collected through a closed-ended questionnaire based on a five-multi choice points. A total of 125 respondents, aged 18 years and above, were selected through convenience sampling. The electronic survey form was prepared using Google Survey and distributed via WhatsApp. Prior to distribution, the questionnaire underwent approval by the supervisor, Dr. Ammad Zafar, following a pilot study. Consent from respondents was obtained prior to their participation in the survey, and ethical considerations were carefully observed throughout the research process.

Secondary data was collected through a review of relevant research papers. Statistical Package for the Social Sciences (SPSS) software was utilized for data analysis. Statistical analysis of the data was conducted using various tools including frequency analysis, cross-tabulation, chi-square test analysis, independent t-test, and binary logistic regression analysis.

ANALYSIS AND INTERPRETATION

Table 01: Age of Respondents

Responses	Percent
16 – 25	37.6
25 – 35	32.0
36 – 45	19.2
45 and above	11.2
Total	100.0

The provided data illustrates the distribution of responses, Specifically, 37.6% of responses fell within the age span of 16 – 25 and 32.0% fell within the range of 26 – 35, other 19.2% of responses were in 36 - 45, additionally 11.2% were 46 - above. This distribution accounts for all responses, totaling 100.0%

Table 02: Proportion of Gender

Gender	Percent
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Male	51.2
Female	48.8
Total	100.0

Gender distribution: 51.2% male and 48.8% female, indicating a relatively balanced representation.

Table 03: Level of Education

Education	Percent
Matriculation	12.8
Intermediate	15.2
Bachelors	55.2
Masters	16.0
PhD	0.8
Total	100.0

Educational achievements are 12.8% with matriculation, 15.2% for intermediate education, 55.2% holding bachelor's degrees, 16.0% with master's degrees and 0.8% with PhD.

Table 04: Mapping route: primary mode of travelling

Responses	Percent
Car	38.4
Walking	19.2
Public transport	8.8
motorcycle	32.8
bicycle	0.8
Total	100.0

This indicates 38.4% commuters travel through cars, while 19.2% travel through public transport and 8.8% prefers walking, other 32.8% uses motorcycles for their daily travel and just 0.8% uses bicycles for their routine travels.

Table 05: Time Travelers: time spend on travelling each day

Responses	Percent
Less than 15 mins	8.8
15 to 30 mins	25.6
30 mins to 1 hr.	34.4
1 to 2 hr.	22.4
2 hr. and above	8.8
Total	100.0

Out of 100% total 8.8% people take less than 15 mins while travelling, around 25.6% reaches in about 15 to 30 mins to their destination and 34.4% people said it take 30 mis to 1 hr. daily during

travelling, 22.4% responded that they are engaged for about 1 to 2 hr. daily while commuting and 8.8% got more than 2 hr. time to spend on travelling each day.

Table 06: Destination Discovery: rating the convenience and accessibility of transportation options

Responses	Percent
very inconvenient and inaccessible	5.6
inconvenient and inaccessible	20.0
neutral	36.8
convenient and accessible	32.8
Very convenient and accessible	4.8
Total	100.0

Only 5.6% feels that the transportation option in their area is very inconvenient and inaccessible, 20% feels that its inconvenient and inaccessible to get better transport and 36.8% are neutral about the accessibility of transport, some of them around 32.8% founds the transport options convenient and accessible and finally we have 4.8% who said its very convenient and accessible to find daily transport.

Table 07: Mental Health Milestones: transportation options impact ability to access mental health services or support

Responses	Percent
Not at all	12.8
Slightly	26.4
Moderately	39.2
Very much	14.4
Extremely	7.2
Total	100.0

Few of them like 12.8% feels transportation does not impact their mental health at all. Around 26.4% feels it slightly impacts mental health and 39.2% feels it moderately impacts their mental wellbeing, 14.4% people are those who feels very much stressed and yes, they said transportation effects mental health finally around 7.2% extremely feel so.

Table 08: Appointment Adventures: missed appointments or engagements due to transportation-related issues

Responses	Percent
Never	23.2
Occasionally	28.0
Sometimes	32.0
Often	12.0
Always	4.0

Total	99.2
Missing	0.8

in this table responses we got says that 23.2% never missed and appointment or engagements due to transport related issues, around 28.0% occasionally and 32.0% sometimes missed their appointments due to transport related issues, 12% often misses and 4% always use to miss their scheduled appointments due to transport related issues however 0.8% didn't wanted to answer this question.

Table 9: Safety Signals: safety of the transportation options

Responses	percent
Very unsafe	24.8
Unsafe	30.4
Neutral	25.6
safe	12.8
Very safe	6.4
Total	100.0

Among Our respondents 24.8% feels very unsafe regarding safety of the transportation options and 30.4% thinks transportation available is unsafe, 25.6% people are neutral so they neither feel it unsafe or safe, 12.8%people thinks that its safe and a group of around 6.4% feels it very safe and we get over all 100% responses regarding safety concerns.

Table 10: Well-being Wheels: better transportation options would improve overall mental well-being?

Responses	Percent
Strongly disagree	23.2
Disagree	25.6
Neutral	36.8
agree	13.6
Strongly agree	0.8
Total	100.0

Around 23.2% strongly disagreed that transportation option would not improve mental wellbeing, 25.6% again disagreed that better transport option would not affect on mental wellbeing 36.8% are neutral about this matter and 13.6% agreed that having better transportation options would improve your overall mental health among these .8%strongly agreed regarding the statement.

Table 11: Reliability Roadmap: reliability of transportation

Responses	Percent
Very unreliable	8.0
Unreliable	18.4
Neutral	38.4
reliable	30.0

Very reliable	4.8
Total	100.0

8.0% feels current transportation options are very unreliable, 18.4% thinks yes current transportation options are unreliable among them 38.4% are neutrals and 30% responded reliable transport options we have just 4.8% feels our current transportation options as we have been very reliable.

Table 12: Stress Stop: experiencing stress or anxiety related to commute

Responses	Percent
Occasionally	8.0
sometimes	16.8
often	42.4
always	32.8
Total	100.0

Around 8.0% occasionally experience stress related to their commute, 16.8% of them sometimes and 42.4% often felt stress or anxiety related to the commute, 32.8% responded that they always feel stressed related to the commute.

Table 13: Social Crossroads: socializing or interaction with others during commute

Responses	Percent
Never	35.2
Occasionally	15.2
sometimes	30.4
often	16.8
always	2.4
Total	100.0

Social interaction was asked by us and 35.2% never interacts with others, while 15.2% occasionally and 30.4% sometimes interacts with others during travelling. 16.8% responded often and 2.4% always interacts while they travel.

Table 14: Loneliness Lanes: feelings of isolation or loneliness related to transportation barriers

Responses	Percent
Never	1.6
Occasionally	15.2
Sometimes	33.6
often	44.0
always	5.6
Total	100.0

Loneliness or feeling of isolation while travelling was our question and respondents responded that 1.6% of them never felt lonely during travel, 15.2% responded that occasionally they feel

isolated and 33.6% among them were those who sometimes felt lonely or isolated, 44% responded often they feel lonely and 5.6% of them always feel lonely or isolated while they travel.

Table 15: Backup Routes: access to alternative transportation

Responses	percent
Not at all	10.4
Slightly	22.4
Moderately	36.0
Very much	28.0
extremely	3.2
Total	100.0

In case primary mode of transportation is unavailable 10.4% do not have access to alternative transport, 22.4% have slight chances of alternate transport and 36.0% have moderate availability of alternative transport, 28.0% responded they very much have alternate transportation availability and 3.2% responded extremely that yes, they have alternative options.

Table 16: Fatigue Freeway: feeling overwhelmed or fatigued by commute

Responses	Percent
Never	15.2
Occasionally	15.2
Sometimes	44.0
often	14.4
always	11.2
Total	100.0

Respondents who were never fatigued by their commute were 15.2% and again 15.2% occasionally felt fatigued due to their commuting experience, 44.0% answered sometimes they feel overwhelmed and fatigued by the commute and 14.4% said often they are fatigued among them those who always feel fatigued were 11.2%.

Table 17: Fitness Freeway: transportation options impact ability to engage in physical activity or exercise

Responses	Percent
Not at all	15.2
Slightly	24.0
Moderately	21.6
Very much	31.2
extremely	8.0
Total	100.0

Among all the respondents 15.2% responded they not at all engage in any physical activity while they travel, 24% of them slightly felt that transportation options impact your ability to engage in

physical activity.21.6% shows moderately it impacts physical activity and 31.2% show that very much it impacts physical activity and exercise finally 8.0% feels it extremely impacts.

Table 18: Satisfaction Station: satisfied with current transportation situation, and how it affects mental health?

Responses	Percent
Very dissatisfied, negatively effects on mental health	4.8
Dissatisfied negatively effects mental health	14.4
Neutral	52.0
Satisfied, positively effects mental health	20.0
Very satisfied, positively effects mental health	8.8
Total	100.0

In the last we had a satisfaction scale regarding current transport conditions and its effects on mental health and 4.8% responded yes transportation situation is very dissatisfied, and it negatively impacts mental health,14.4% are dissatisfied, while 52% are neutral and 20% responded they are satisfied with current transport situation finally 8.8% were very satisfied and it positively impacts their mental health, all these responses are taken.

Table 19: Independent variable daily commuting style

Responses	Percent
No	22.4
Yes	77.6
Total	100.0

77.6% respondents were in favor of yes that transportation actually effects mental wellbeing and 22.4% didn't felt that mental wellbeing is related through transportation situations.

Table 20: dependent variable psychological well being

Responses	Percent
No	8.0
Yes	92.0
Total	100.0

It feels 92.0% mental wellbeing is affected by transportation situations and 8.0% population responded no mental wellbeing is not affected by transport.

CROSSTAB ANALYSIS

Table 21: cross tabulation analysis between daily commuting and mental well being

		Daily commuting		
		no	yes	total
Psychological wellbeing	no	50%	50%	100%
	yes	20%	80%	100%
	total	22.4%	77.6%	100%

Table 21 likely displays the percentage that how much is daily commuting and transportation is related to psychological wellbeing.

CHI-SQUARE ANALYSIS

Table 22: Chi-Square Analysis between daily commuting and psychological well being

	Value	d f	Asymptotic Significance (2- sided)	Exact Sig. (2- sided)	Exact Sig. (1- sided)
Pearson Chi-Square	4.763 ^a	1	0.029	0.044	0.044
Continuity Correction ^b	3.194	1	0.074		
Likelihood Ratio	4.026	1	0.045		
Fisher's Exact Test					
Linear-by-Linear Association	4.725	1	0.030		
N of Valid Cases	125				

a. 1 cells (25.0%) have expected count less than 5. The minimum expected count is 2.24.

b. Computed only for a 2x2 table

In table 22 pearson Chi-Square value is 4.763 with 1 degree of freedom, yielding an asymptotic significance of .029. Continuity Correction yields a value of 3.194 with a significance of 0.074. The Likelihood Ratio gives a value of 4.026 with a significance of .045 Fisher's Exact Test has a significance of 0.044 (two-sided) and .044(one-sided). The Linear-by-Linear Association is 4.725 and it shows a significance of .030. The analysis is based on 125 valid cases. Note that 1 cell have an expected count less than 5, with the minimum expected count being 2.24. Continuity Correction is computed only for a 2x2 table.

INDEPENDENT T-TEST ANALYSIS

Table 23: Independent T-Test

	Gender	N	Mean	Std. Deviation	Std. Error Mean
Daily commuting	male	64	0.7656	0.42696	0.05337
	female	61	0.7869	0.41291	0.05287
Mental well being	male	64	0.9688	0.17537	0.02192
	female	61	0.8689	0.34036	0.04358

in table 23, For daily commuting, among males (N=64), the mean value is 0.7656. with a standard deviation of 0.42696 and a standard error of the mean of 0.05337. Among females (N=61), the mean value is 0.7869. with a standard deviation of 0.41291 and a standard error of the mean of 0.05287. Concerning mental wellbeing among males (N=64), the mean value is 0.9688. with a standard deviation of 0.17537 and a standard error of the mean of 0.02192. Among females (N=61), the mean value is at 0.9688. with a standard deviation of 0.34036 and a standard error of the mean of 0.04358.

Table 24: Independent t test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2- tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
IND	Equal variances assumed	0.321	.572	-.283	123	.778	-.02126	.07518	-.17008	.12756
	Equal variances not assumed			-.283	122.973	.778	-.02126	.07512	-.16996	.12744
DV	Equal variances assumed	19.612	.000	2.077	123	.040	.09990	.04810	.00468	.19511
	Equal variances not assumed			2.048	88.791	.044	.09990	.04878	.00297	.19683

The analysis conducted includes Levene's Test for Equality of Variances and T-tests for Equality of Means. For the daily commuting experience when equal variances were assumed, the Levene's test yielded F (.321), sig (.572), and the t-test indicated t (-.283), sig2tailed (.778), with a mean difference of (-.02126) and a 95% confidence interval ranging from (-.17008 to .12756). When equal variances were not assumed, the t-test results were similar, showing t (-.283) F (.321), sig (.572), and t-test indicated t(-.283), with a mean difference of (.02126) and a 95% confidence interval ranging from (-.16996 to .12744) For psychological wellbeing outcome, assuming equal variances, Levene's test indicated F (19.612), sig (000), and the t-test showed t(2.077), sig2tailed. (040), with a mean difference of (.09990) and a 95% confidence interval ranging from (.00468 to .19511) . When equal variances were not assumed, the t-test results showed t (2.048), sig2tailed (.044), with a mean difference of (.09990) and a 95% confidence interval ranging from (.00297 to .19683).

BINARY LOGISTIC REGRESSION ANALYSIS

Table 25: binary logistic regression analysis on daily commuting effects mental well being

	B	S.E.	Wald	df	Sig.	Exp(B)
Step 1 ^a Daily commuting	1.386	.674	4.230	1	.040	4.000
	1.526	.493	9.565	1	.002	4.600

a. Variable(s) entered on step 1: IND.

In Step 1 of the analysis, the variable daily commuting was entered. The coefficient (B) associated with VP was 1.386, with a standard error (S.E.) of .674. The Wald statistic was 4.230, with 1 degree of freedom (df), and a significance level (Sig.) of .040. The odds ratio (Exp(B)) for VP was calculated to be 4.000. Additionally, a constant term was included with a coefficient of 1.526, a standard error of .493, a Wald statistic of 9.565, and a significance level of 0.002. The odds ratio for the constant was 4.600.

DISCUSSION

The discussion section of this research provides a synthesis of the key findings and offers insights, interpretations, and implications for understanding the psychological puzzle of daily commuting. Here's a structured discussion based on the themes identified

1. Impact of Commute Duration:

The consistent evidence linking longer commute durations with negative mental health outcomes underscores the significance of commute duration as a critical factor influencing commuters' well-being. Commuters facing prolonged travel times may experience heightened stress, fatigue, and diminished life satisfaction. This suggests the need for interventions aimed at reducing commute times through measures such as improved transportation infrastructure.

2. Mode of Transportation:

The contrasting effects of different transportation modes highlight the importance of mode choice in shaping commuters' experiences and mental well-being. While public transit commuting is associated with higher stress levels due to factors like crowding and delays, active modes such as walking and cycling offer potential benefits for mental health. This suggests the importance of promoting active transportation options and enhancing public transit services to accommodate commuters' diverse needs and preferences.

3. Coping Strategies:

The identification of various coping mechanisms underscores the resilience and adaptability of commuters in managing the challenges of daily commuting. Effective coping strategies, such as mindfulness practices, social interaction, and engaging in enjoyable activities during the commute, play a crucial role in mitigating the negative effects of commuting on mental well-being. Interventions aimed at promoting these coping strategies could enhance commuters' resilience and overall well-being.

4. Environmental and Contextual Factors:

Environmental stressors and contextual factors interact with commuting experiences to influence commuters' perceptions and well-being. Addressing environmental factors such as noise, air pollution, and urban design can contribute to a more positive commuting environment and reduce stress levels among commuters. Additionally, fostering supportive work environments and social networks may buffer the negative effects of commuting on mental well-being and promote overall resilience.

CONCLUSION

In conclusion, this research emphasizes the need to recognize commuting as a multidimensional phenomenon with profound implications for individuals' mental health. By understanding the psychological dynamics of commuting and implementing targeted interventions, policymakers, urban planners, and employers can create healthier, more sustainable commuting environments that enhance

overall well-being for all commuters. Continued research and collaborative efforts are essential for advancing our understanding and addressing the complex challenges posed by daily commuting.

RECOMMENDATION

based on the findings and insights derived from this research on the psychological puzzle of daily commuting, several recommendations can be made to enhance commuters' mental well-being and improve the overall commuting experience: Invest in

Transportation Infrastructure:

Governments and transportation authorities should prioritize investments in transportation infrastructure to reduce commute times and improve the reliability and efficiency of public transit systems. Initiatives such as expanding public transit networks, implementing dedicated bus lanes, and enhancing bicycle and pedestrian infrastructure can provide commuters with more reliable and sustainable transportation options.

Promote Active Transportation:

Encourage the adoption of active transportation modes such as walking and cycling by investing in infrastructure that supports safe and convenient active travel. Creating pedestrian-friendly urban environments, building bike lanes, and providing secure bike parking facilities can incentivize commuters to choose active modes of transportation, leading to improvements in both physical and mental well-being.

Promote Mental Health Awareness: Increase:

awareness of the psychological impact of commuting and provide resources and support for commuters to manage stress and enhance resilience. Employers can offer employee assistance programs (EAPs) that provide access to counseling services, stress management workshops, and mindfulness training. Additionally, public health campaigns and community initiatives can raise awareness of the importance of mental well-being and promote self-care practices among commuters.

Address Socioeconomic Disparities:

Take proactive measures to address transportation equity issues and reduce disparities in commuting experiences and mental health outcomes. This includes ensuring equitable access to affordable and reliable transportation options for all communities, especially those in underserved areas. Subsidized transit fares, targeted transportation subsidies for low-income individuals, and community-based transportation initiatives can help alleviate financial barriers to commuting.

Improve Environmental Quality:

Mitigate environmental stressors such as noise and air pollution to create a more pleasant and healthy commuting environment. Implement measures to reduce vehicle emissions, promote green spaces along commuting routes, and create pedestrian-friendly urban environments to enhance the overall quality of the commuting experience and support commuters' mental well-being. Research and Evaluation: Continue conducting research to further understand the psychological aspects of commuting and evaluate the effectiveness of interventions aimed at promoting mental well-being among commuters. Longitudinal studies tracking changes in commuting patterns and mental health outcomes over time can provide valuable insights into the long-term impacts of interventions and guide future policy and programmatic efforts. By implementing these recommendations, stakeholders can work together to create a commuting environment that supports commuters' mental well-being, enhances overall quality of life, and contributes to the development of healthier and more sustainable communities.

LIMITATION OF RESEARCH

While this research provides valuable insights into the psychological puzzle of daily commuting, it is important to acknowledge several limitations that may affect the interpretation and generalizability of the findings:

1. **Sampling Bias:** The study may be subject to sampling bias, as participants were recruited through convenience sampling methods, such as online surveys. This could limit the representativeness of the sample and introduce biases related to demographics, commuting patterns, and other factors.
2. **Self-Report Measures:** The reliance on self-report measures for collecting data on commuting experiences and mental well-being may introduce response bias and social desirability bias. Participants may underreport negative experiences or overreport positive experiences, leading to inaccuracies in the data.
3. **Cross-Sectional Design:** The use of a cross-sectional research design limits the ability to establish causal relationships between commuting experiences and mental well-being outcomes. Longitudinal studies would be needed to track changes in commuters' mental health over time and identify causal pathways.
4. **Limited Scope of Variables:** The study may not have captured all relevant variables that could influence commuting experiences and mental well-being outcomes. Factors such as individual differences in coping strategies, social support networks, and environmental conditions may not have been adequately accounted for in the analysis.
5. **Generalizability:** The findings of the research may not be generalizable to all populations or contexts, as they are based on data collected from a specific sample of commuters in a particular geographic area or cultural context. Cultural differences, variations in transportation infrastructure could influence the applicability of the findings to other settings.
6. **Response Rate and Attrition:** The study may have experienced low response rates or attrition, particularly in longitudinal studies, which could introduce bias and affect the reliability and validity of the results.
7. **Measurement Validity and Reliability:** The validity and reliability of the measurement instruments used to assess commuting experiences and mental well-being may be subject to limitations. The use of standardized scales and validated measures can help mitigate these concerns, but measurement error and instrument limitations should be considered.

Despite these limitations, this research contributes valuable insights into the psychological dimensions of daily commuting and lays the groundwork for future studies to address these limitations and further advance our understanding of commuting experiences and mental well-being.

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